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MILITARY SPECIFICATION

POWER SUPPLY, D.C., FOR OXYGEN GENERATOR

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers direct current (dc) power supplies designed to furnish electrical power to submarine electrolytic oxygen generators.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-14	- Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-R-19	- Resistors, Variable, Wire-Wound (Low Operating Temperature), General Specification for.
MIL-B-117	- Bags, Sleeves and Tubing - Interior Packaging.
MIL-S-901	- Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-C-2212 - Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-L-3661 - Lampholders, Indicator Lights, Indicator-Light Housings, and Indicator-Light Lenses, General Specification for.
MIL-S-3786 - Switches, Rotary (Circuit Selector, Low-Current Capacity), General Specification for.
MIL-S-3786/4 - Switch, Rotary, Closed Construction, 2 Amperes, Style SR04.
MIL-C-5015 - Connectors, Electric, Circular Threaded, AN Type, General Specification for.
MIL-S-8805 - Switches and Switch Assemblies, Sensitive, and Push (Snap Action), General Specification for.
MIL-S-8805/23 - Switch Assemblies, Sensitive, Push, Momentary, 5 Amperes, Unsealed.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements for.
MIL-F-15160 - Fuses: Instrument, Power, and Telephone.
MIL-F-15160/60 - Fuses: Instrument, Power, and Telephone (Nonindicating), Style F60.
MIL-S-15291 - Switches, Rotary, Snap Action and Detent/Spring Return Action, General Specification for.
MIL-P-15736 - Power Supplies, Metallic Rectifier, Naval Shipboard.
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-C-17361 - Circuit Breakers, Air, Electric, Insulated Housing (Shipboard Use).
MIL-F-19207 - Fuseholders, Extractor Post Type, Blown Fuse Indicating and Nonindicating, General Specification for.
MIL-F-19207/24 - Fuseholders, Extractor Post Type, Blow Fuse Indicating, Type FHL35W.
MIL-F-19207/25 - Fuseholders, Extractor Post Type, Nonindicating, Type FHN41WB.
MIL-P-24441 - Paint, Epoxy-Polyamide, General Specification for.
MIL-P-24441/6 - Paint, Epoxy-Polyamide, Exterior Topcoat, Dark Gray, Formula 155-RO = 6, Type 1.
MIL-H-24520 - Hose and Hose Assembly for Water Cooling of Electronic Equipment.
MIL-P-55110 - Printed-Wiring Boards, General Specification for.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-275 - Printed Wiring for Electronic Equipment.

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MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-470 - Maintainability Program for Systems and Equipment.
MIL-STD-471 - Maintainability Verification/Demonstration/Evaluation.
MIL-STD-721 - Definitions of Terms for Reliability and Maintainability.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.
MIL-STD-882 - System Safety Program Requirements.
MIL-STD-965 - Parts Control Program.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electric Power, Alternating Current. (Metric)
DOD-STD-1399, Section 532 - Interface Standard for Shipboard Systems - Cooling Water for Support of Electronic Equipment. (Metric)

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.
MIL-HDBK-472 - Maintainability Prediction.

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

S9515-AA-MMO-010 - 6L16/Oxygen, Volume 1, Electrolytic Oxygen Generator 6L16 All Models, Description and Operation, Sections 1-4.
S9515-AA-MMO-020 - 6L16/Oxygen, Volume 2, Parts 1 and 2, Electrolytic Oxygen Generator 6L16 All Models, Troubleshooting and Repairs, Section 5.
S9515-AA-MMO-030 - 6L16/Oxygen, Volume 3, Electrolytic Oxygen Generator 6L16 All Models Corrective Maintenance and Illustrated Parts Breakdown, Sections 6 and 7.
S9515-AA-MMO-040 - Twelve Pulse Hexagon Power Supply for Electrolytic Oxygen Generator 6L16 (All Models).
0967-LP-312-8010 - Maintainability Design Criteria Handbook for Designers of Shipboard Electronic Equipment.
0967-LP-597-1011 - Parts Application and Reliability Information Manual for Navy Electronic Equipment.

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(Copies of specifications, standards, handbooks and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN IRON AND STEEL INSTITUTE (AISI)
Steel Products Manual

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 16th Street NW, Washington, DC 20036.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Power supply requirements. The power supply shall minimize the distortion of the input voltage, up to and including the three hundred thirty-third harmonic of the input frequency, and in no case shall exceed the values specified herein. Power supply shall supply power to the oxygen generator for control power purposes and electrolysis power (350 to 1250 amperes dc (Adc) as described in NAVSEA S9515-AA-MMO-010, NAVSEA S9515-AA-MMO-020, NAVSEA S9515-AA-MMO-030, and NAVSEA S9515-AA-MMO-040. The power supply shall be physically interchangeable with the presently installed power supply. Interchangeability shall be maintained with respect to the following features of power supply design and construction (as described in NAVSEA S9515-AA-MMO-010, NAVSEA S9515-AA-MMO-020, NAVSEA S9515-AA-MMO-030 and NAVSEA S9515-AA-MMO-040):

(a) Overall height, width, and depth of enclosure.
(b) Base mounting holes and mounting centers at bedplate of oxygen generator.
(c) Provisions for alternating current (ac) power and control connections, including input terminal boards, selection and positioning of connector plugs and receptacles, and location of cable entrance openings.

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(d) Size, material, and position of positive and negative dc bus bars extending from power supply.
(e) Material, size, threading, and location of inlet and outlet tubing required for water cooling of equipment (see 3.3.2) and pipe fittings and drains.
(f) Instrumentation, indicators, alarms, and protective shutdown provisions, unless otherwise specified herein.
(g) Methods of energization, control, and protective shutdown, unless otherwise specified herein.
(h) Cover panel bolt, thread type and size.
(i) Location, type, and size of mounting holes on top of the power supply.
(j) The outside enclosures (for example, the frame, cover panels, and so forth) shall be constructed of a material which will not be degraded by potassium hydroxide (KOH) or boiler chemical solutions.

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.2 Table I lists controls, indicators, and protective devices which are necessary to be located on the front panel of the power supply and which shall be located on the front panel of equipment specified herein.

TABLE I. Controls, indicators, and protective devices.

Description	Applicable reference	Remarks
Main circuit breaker	See 3.2.2.1	Electrical, mechanical design, control and safety circuits shall not change
Overload reset	MIL-C-2212	---
Fuseholders	See 3.2.2.3	---
Fuses	See 3.2.2.3	---
Elapsed time meter	MIL-E-917	---
Kiloampere-hour meter	See 3.2.2.6	White
Alternate - normal control power selector switch	See 3.3.5.4	---

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TABLE I. Controls, indicators, and protective devices. - Continued

Description	Applicable reference	Remarks
Motor operated circuit interrupter	See 3.2.2.2	---
Blown fuse indicators for output filter	See 3.3.7	Red
High ripple warning indicator light	See 3.3.7	Red
Low cooling water flow warning indicator light	See 3.3.7	Amber
Indicating lamp - "control power on"	MIL-L-3661	White
Indicating lamp - "dc power on"	MIL-L-3661	White
Rheostat, constant current setting	MIL-R-19	---
Output multi-readout meter	See 3.2.2.4	---
Output multi-readout meter selector switch	See 3.3.5.1	---
Output multi-readout meter indicator lampcheck switch	See 3.3.5.2	---
Ground detector momentary pushbutton switch	See 3.3.5.3	---
Indicating lamps	See 3.3.7.1	Green

3.2.2.1 Main circuit breaker. The main circuit breaker shall be type AQB-A250 in accordance with MIL-C-17361, and as follows:

(a) Voltage and frequency ratings: 500 volts, 60 hertz (Hz).
(b) Number of poles: three.
(c) Connections: back-connected with removable mounting.
(d) Trip element rating: 175N, as specified in MIL-C-17361.

3.2.2.2 Motor operated circuit interrupter. The motor operated circuit interrupter shall be type NQB-A250 in accordance with MIL-C-17361 and as follows:

(a) Voltage and frequency ratings: 500 volts, 60 Hz.
(b) Number of poles: three.
(c) Connections: front-connected.

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(d) Trip element rating: 175N, as specified by MIL-C-17361.
(e) Operating mechanism: electric motor operated, with 115 volt, 60 Hz input to motor.
(f) Undervoltage trip device shall have 115 volt, 60 Hz input.
(g) Auxiliary switches shall consist of single pole double throw auxiliary switch and two circuit double break auxiliary switches.

3.2.2.3 Control fuses. Control fuses shall be characteristic C in accordance with MIL-F-15160 and MIL-F-15160/60. Fuseholders with blown fuse indicator in accordance with MIL-F-19207, MIL-F-19207/24 and MIL-F-19207/25 shall be provided for control fuses.

3.2.2.4 Output multi-readout meter. The output multi-readout meter shall be digital, bipolar, 3.5 digital display and shall have a tolerance of 0.05 percent of full-scale, plus or minus one digit.

3.2.2.4.1 The total error when indicating output current shall not exceed plus or minus 1.15 percent of full power supply current output. This shall include shunt, circuitry, and the meter.

3.2.2.4.2 The shunt shall be water cooled as described in NAVSEA S9515-AA-MMC-040 and shall be a 1000 ampere per 100 millivolt (mV) shunt.

3.2.2.5 Printed circuit cards. Printed circuit cards shall contain diagnostics and light emitting diodes (LEDs) as described in NAVSEA S9515-AA-MMO-040. Printed circuit cards shall be wired and printed in accordance with MIL-P-55110.

3.2.2.6 Kiloampere-hour meter. The kiloampere-hour meter shall provide a five-digit read-out extending to a full-scale reading of 99,999 kiloampere-hours. Kiloampere-hour indicator shall be an electromechanical type displaying five digits. The counter digits shall not be smaller than 5/32 inch high and 1/8 inch wide and shall be white on a black background. The bezel of the indicators shall be constructed of metal or plastic material. The metal shall be of either corrosion resistant material or material treated for corrosion resistance in accordance with requirement 51 of MIL-STD-454. The plastic material shall be type MMI, MAI, MSG, MSI, or SDG-F in accordance with MIL-M-14. A reset device operated with a special tool shall be provided for the digital counter to enable resetting of the indicator in place. Kiloampere-hour metering system shall be indicative of the output current of the electrolytic oxygen generator and display the cumulative total by means of the five-digit counter (in kiloampere hours). The system shall be designed to operate from a 1000 ampere per 100 mV shunt. The system shall be calibrated so that a current flow of 1000 amperes for 1 hour will cause the counter to advance one unit. Accuracy of the indication for current ranges shall be 200 to 1200 amperes ± 1 percent of full-scale. Means shall be provided for the calibration of the metering system.

3.2.2.7 Elapsed time indicator. The elapsed time indicator shall provide a full-scale read-out extending to 99,999 hours in 1-hour steps.

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3.2.3 Electrical and mechanical components and design features. Electrical and mechanical components and design features listed in table II shall not change due to interfaces which shall be maintained to permit installation of equipment in the oxygen generator, as shown in NAVSEA S9515-AA-MMO-030 and NAVSEA S9515-AA-MMO-040.

TABLE II. Electrical and mechanical components and design features.

Description	Interface to be maintained
Control power transformer, single phase, 1.12 kVA, 440 volts primary, 125 volts secondary, 60 Hz, core type AISI-M6 silicon steel 3.43 square inches	Electrical
Transformer for control cubicle, 1.21 kVA, 440 volts primary, 244 volts secondary, 60 Hz, single phase, core AISI-M5 silicon steel 2.85 square inches	Electrical
Transformer for control cubicle, 3-phase, 0.7 kVA, 440 volts primary, 75.4 volts secondary, 60 Hz, core AISI-M6 silicon steel 2.14 square inches	Electrical
Power plug, cubicle interconnection, MIL-C-5015, straight plug, series QWLD, bright nickel finish, MS insert No. 26-2S, 0.0001 gold plated crimp type pin contacts	Mechanical, electrical, design and physical location
Power plug, cubicle interconnection, MIL-C-5015, straight plug, series QWLD, bright nickel finish, MS insert No. 22-18P, 0.0001 gold plated solder type pin contacts	Mechanical, electrical, design and physical location
Positive and negative rectifier output, copper bus bar terminals	Material mechanical configuration and location
Incoming ac cable entrance point	Physical configuration and location, entrance point
bulkhead unions for "Water in" and "Water out"	Mechanical design and location
Welded frame assembly	Height, width, depth, mounting holes top, sides, and bottom mounting holes; and the capability to lift and transport the power supply using a lifting device attached to the top mounting holes

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TABLE II. Electrical and mechanical components and design features. - Continued

Description	Interface to be maintained
Alternate power cable terminal block	Physical location near cable entrance
Rectifier high temperature shutdown switch	Device 2TT, single throw, double pole, thermal switch. The switch, when activated, shall provide a shutdown and a backup shutdown signal to the oxygen generator electronic control cubicle. The device shall change state in an equivalent time as that of device 2TT.

3.2.3.1 Floating nuts. Floating nuts may be used at the interface between power supply and control cubicle as a means to attach the power supply to the control cubicle of the oxygen generator.

3.2.4 Heat shields. Heat shields of an asbestos free material shall be bolted to the rear panels of the power supply. Johns Mansville asbestos free insulation material, or equivalent shall be used.

3.2.5 Epoxy base paint. Epoxy base paint, in accordance with MIL-P-24441 and MIL-P-24441/6 shall be applied to enclosure frame and sheet metal surfaces of the power supply.

3.2.6 Enclosure. Power supply shall be furnished with a dripproof protected enclosure, as defined in MIL-STD-108. Power supply enclosure shall exclude drops of liquid or solid particles striking the enclosure at an angle of 60 degrees from the vertical. Enclosures shall be of the dead-front type. Ventilation louvers shall be restricted to front and right-hand surfaces of the power supply enclosure. Weight of the power supply shall not exceed 1575 pounds.

3.2.7 Burn-in. The hexagon power supply shall withstand 504 hours operating at 1250 Adc under a simulated oxygen generator load.

3.2.8 Endurance. The hexagon power supply shall furnish power to the electrolytic cells of the oxygen generator for a period of 336 hours while operating at full load.

3.3 Design. Power supply shall be completely static and shall conform to the safety requirements of MIL-E-917. Equipment shall also incorporate design and construction techniques which provide maximum safety to operating and maintenance personnel in accordance with requirement 1 of MIL-STD-454. Electrical insulation materials used in construction of the power supply equipment

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shall be processed in accordance with insulating procedures specified in MIL-E-917. Electrical insulation shall be class B, F, H, or N in accordance with MIL-E-917. Selection and application of electrical piece parts shall be as specified in MIL-E-917.

3.3.1 Circuitry. Power supply shall have a transformer input so that the dc output circuit is isolated from the ac input. Transformers shall have electrically isolated windings (primary and secondary). Neither autotransformers nor open-delta connected transformers shall be used at the ac line input of the power supply. Fast acting current limiting fuses may be used where required for power rectification circuitry.

3.3.2 Cooling. Power supply shall be water cooled, and shall conform to the requirements of DOD-STD-1399, section 532. Power supply shall operate in a shipboard space with an ambient temperature range of 7 to 50 Celsius (°C). Cooling water shall be supplied to the power supply at a flow rate of 5 gallons per minute (gal/min) and a temperature of 35°C. Pressure drop for cooling water, from inlet to outlet of the power supply shall not exceed 20 pounds per square inch (lb/in^2) at 5 gal/min flow rate. Cooling water shall be returned at a temperature not to exceed 12°C above inlet water temperature. When cooling water is supplied at specified temperature and flow rate, outlet water temperature shall not exceed 47°C. In the event that flow of cooling water decreases to 2.5 gal/min flow rate, a "water flow warning" indicator shall be actuated. Provision shall be made for protective shutdown of the power supply when the temperature of any power semiconductor increases to the maximum case temperature rating of the device. The protective shutdown device shall be an equivalent circuit of device 2TT in NAVSEA S9515-AA-MMO-040.

3.3.2.1 Design pressure. Power supply cooling system shall withstand 120 lb/in^2 pressure at cooling water inlet.

3.3.2.2 Materials. Copper alloys shall be used in the fabrication of rigid tubing and water jackets which carry coolant within the power supply. Hoses and hose assemblies for cooling water, and end fittings for cooling water hoses, shall be in accordance with MIL-H-24520. Materials used within the cooling water system shall not degrade the purity of inlet water which is supplied at resistivity of 100,000 ohm-centimeter. Conductivity of water shall be 2.0 microsiemens per centimeter in accordance with DOD-STD-1399, section 532.

3.3.3 Noise reduction. The preferred method of equipment noise reduction shall be implemented by design and construction of circuits and components used therein. Where components are mounted in the power supply assembly by resilient mounting means (sound isolators), exposed metal parts of such components (for example, transformer cores) shall be connected to the equipment frame by a grounding lead to reduce potential electric shock hazard.

3.3.4 Access. Fuses shall be accessible through the front of the power supply without removal of any other components. Parts shall be accessible for removal through the front or right hand side of the power supply, when facing the front only, except for the current shunt (A4R1) and choke (A4L1) which require access from the rear.

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3.3.5 Switches.

3.3.5.1 Output multi-readout meter selector switch. The output multi-readout meter selector switch shall be four-position, 1.0 ampere break before make, contact switch, construction E, characteristic B, style SR04-1, number 0058, as specified in MIL-S-3786 and MIL-S-3786/4. The four positions are for four separate functions and shall be wired accordingly with the output multi-readout meter. The four functions shall be as follows:

(a) Output amperage, from 0 to 1500 Adc.
(b) Output voltage, from 00.0 volts direct current (Vdc) to positive 80.0 Vdc.
(c) Positive voltage to ground, from 00.0 to positive 60.0 Vdc.
(d) Negative voltage to ground, from 00.0 to negative 60.0 Vdc.

3.3.5.1.1 Voltage to ground positions. Positions (c) and (d) of 3.3.5.1 of the output multi-readout meter selector switch shall connect a 100 ohm, 25 watt, 1 percent resistor from the power supply frame to either the negative or positive power supply output bus. This resistor shall only be connected while in these switch positions. These two switch positions shall also connect the output multi-readout meter (see 3.2.2.4) across the 100 ohm, 25 watt, 1 percent resistor, to indicate cell bus voltage to ground.

3.3.5.2 Output multi-readout meter indicator lampcheck switch. A switch shall be located on the front panel of the power supply and shall be used in conjunction with the output multi-readout meter such that when the switch is actuated, it shall test all indicator internal lamp segments of the meter. The switch shall be a pushbutton type and shall not be capable of sustaining energization unless held depressed by the equipment operator.

3.3.5.3 Ground detector momentary pushbutton switch. The ground detector momentary pushbutton switch shall be in accordance with MIL-S-8805 and number 001 of MIL-S-8805/23.

3.3.5.4 Terminal strip end alternate-normal control power selector switch. A terminal strip shall be included within the power supply to provide a 440 Vac line as alternate power source to the control cubicle of the oxygen generator. The power supply shall incorporate a control power selector switch which permits the control power of the oxygen generator to be manually transferred from normal to alternate 440-volt line input. The switch shall be in accordance with MIL-S-15291 and shall consist of normal, alternate, and off positions. The control power selector shall be mounted on the front surface of the power supply.

3.3.6 Terminal boards. Terminal boards within the power supply shall be vertically mounted in an orientation to prevent entrapment of moisture across electrical connections.

3.3.7 Indicators. Power supply shall incorporate indicators as specified below:

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(a) High ripple warning indicator light which is activated when total root mean square (rms) ripple current increases to a value 6 decibels (dB) in excess of the total rms ripple current specified herein.
(b) Blown fuse indicator for output filter capacitor fuses.
(c) Water flow warning indicator light in accordance with MIL-L-3661.

Indicators shall be mounted on the front surface of the power supply. The use of LEDs as indicators for blown capacitor fuses and excessive output ripple current is acceptable.

3.3.7.1 Indicating lamps. The indicating lamps shall have three green LEDs to indicate loss of 15 volts plus, loss of 15 volts minus, and loss of 24 volts plus.

3.3.8 Test points. Test points suitable for insertion of instrument probes shall be provided for troubleshooting of the power supply. Test points shall be grouped in a location which is accessible at the front of the power supply enclosure. Access to the test points shall be obtained through a small door in the front cover plates as shown in NAVSEA S9515-AA-MMO-040.

3.3.9 Weight. The weight of the power supply shall be kept to a minimum. At delivery, the net weight and also the operating weight with fluids, where applicable, shall be indicated. In addition, the vertical, longitudinal, and transverse location of the center of gravity (wet or dry, as appropriate) of each power supply weighing 100 pounds or more, shall be shown on the outline or assembly drawing.

3.4 Operation. The power supply shall operate as specified herein and shall not be adversely affected over the tolerance bands, ranges and limits or any condition of loading imposed by the oxygen generator when supplied with type I power as described in DOD-STD-1399, section 300. The equipment shall be designed for operation on a 3-phase, 3-wire ungrounded power source; input power shall be 440 volts, 3-phase, 60 Hz line input. Direct or capacitive grounding of input lines or internal power supply circuitry shall not be permitted. Current drawn by the equipment shall be balanced to the extent that the current in any wire shall not differ by more than 5 percent from the average current in the three wires. The output of the power supply shall be continuously variable at any set output current level between 0 and 1250 Adc. Output voltage shall be 27 to 55 Vdc. Current regulation shall apply between 350 and 1250 Adc. Automatic current regulation shall be within an accuracy of plus or minus 1 percent of the full rated 1250 Adc when supplying power to an oxygen generator load defined as follows:

(a) Minimum load - 0.01 ohm + 24 Vdc (EMF) at 180 ± 5 degrees Fahrenheit (°F).
(b) Minimum load - 0.029 ohm + 24 Vdc (EMF) at 180 ± 5°F.

3.4.1 The input power factor for the power supply and the kilovoltampere (kVA) is dependent on the load conditions as follows:

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(a) Efficiency at 1250 amperes, R = 0.019 ohm is 81 percent - range 0.80 to 0.83.
(b) Power factor at 1250 amperes, R = 0.019 ohm is 70 percent - range 0.68 to 0.72.
(c) Voltage spike - 1880 - 2658 line to ground.
(d) Plus or minus 10 percent in 2 seconds.

The power supply shall have an efficiency of at least 65 percent. Input voltage transient requirements and voltage spike requirements shall be in accordance with DOD-STD-1399, section 300.

3.5 Harmonic distortion. With the exception of the seventh, eleventh, and thirteenth order harmonics of power frequency, the measured magnitude of each of the individual line harmonic currents, from the second through the twenty-third, expressed as a percentage of the full-load fundamental rms line current, shall not exceed 3 percent for any input current drawn from the 440-volt line, when the power supply is operated at output currents between 350 and 1250 Adc. Harmonic line currents of order seven, eleven, and thirteen, expressed as a percentage of full-load fundamental rms line current, shall not exceed the percentages specified below, for any input current drawn from the 440-volt line, when the power supply is operated at output currents between 350 and 1250 Adc.

(a) 2.25 percent for seventh harmonic.
(b) 6.5 percent for eleventh harmonic.
(c) 4.5 percent for thirteenth harmonic.

Similarly, no individual harmonic line current from twenty-fourth harmonic through 20 kilohertz shall exceed a magnitude in percent equal to $\left(\frac{1700}{n^2}\right)$, where "n" is the harmonic multiple number. Harmonic currents shall be measured as specified in 4.6.3.

3.6 Output ripple. Output ripple current distortion shall meet the following conditions:

(a) Rms current level of individual ripple frequencies shall not exceed 0.2 percent of the dc output maximum current level over the complete output current range.
(b) Maximum total rms ripple current distortion shall not exceed 1.0 percent of dc output maximum current over the complete output current range.

3.7 Airborne noise. Airborne noise generated by the power supply shall not exceed acceptance levels of table III for sound pressure expressed as decibels referred to 20 micropascals.

TABLE III. Acceptance levels for sound pressure.

Octave band center frequency (Hz)								
31.5	63	125	250	500	1000	2000	4000	8000
95	90	85	82	79	76	75	75	75

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3.8 Structureborne noise. Structureborne noise generated by the power supply shall not exceed type 3 acceptance limits of MIL-STD-740 for solidly mounted life support equipment.

3.9 Shock and vibration. The power supply shall meet the shock requirements of MIL-S-901, grade A, class I, type A, and vibration requirements of MIL-STD-167-1, type I.

3.10 Technical data. The contractor shall prepare drawings in accordance with the data ordering documents included in the contract or order (see 6.2.2), and the additional drawing requirements of 3.10.1.1 shall be included.

3.10.1 Technical manuals. Preliminary and standard technical manuals shall be in accordance with type I of MIL-M-15071.

3.10.1.1 Engineering drawings shall be presented as a set, and shall provide data on component selection, equipment installation, and overall power supply design. Each set of engineering drawings shall contain the following technical features:

(a) Installation drawings shall demonstrate interchangeability between existing equipments and the power supply described herein (see 3.2). Drawings shall contain the following information:
  (1) Applicable specification and exceptions.
  (2) Complete rating showing:
    a. Input, voltage, frequency, number of phases.
    b. Input kilowatt (kW), kilovar, total rms line current and fundamental frequency line current; and power factor at zero, 1/2 and full-load.
    c. Output voltage and current at full-load.
  (3) Shock classification.
  (4) Enclosure classification.
  (5) Type of load.
  (6) Ambient temperature range.
  (7) Ripple content of output voltage, and rms ripple component of output current under oxygen generator load.
  (8) Outline drawing of equipment showing front, side, top, and section views as necessary, to present information as follows:
    a. Approximate arrangement and location of components.
    b. Identification and information plates.
    c. Description of design and construction features outlined in 3.2, items (a) through (j).
  (9) Briefly presented operating instructions.
  (10) Pressure drop at flow rate.
  (11) Heat loss in kW or British thermal unit per hour.
(b) Mechanical and electrical schematics, and wiring diagrams containing sufficient information to demonstrate that the configuration of the power supply is consistent with 3.3 herein; drawings shall include the following minimum data:
  (1) Schematic circuit diagram in accordance with MIL-P-15736.
  (2) Connection diagram conforming to MIL-P-15736.
  (3) Description of operation in accordance with MIL-P-15736.

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(c) Detail drawings pertaining to parts, materials, and internal construction sufficient to demonstrate that the equipment satisfies requirements of MIL-E-917 and 3.3 herein, and shall present the following information, prepared in accordance with MIL-P-15736:
(1) List of material.
(2) Transformer and reactor data.
(3) Semiconductor data.

3.10.2 Reliability and maintainability (R and M) program plans. Reliability and maintainability (R and M) program plans shall be prepared in accordance with the data ordering documents included in the contract or order (see 6.2.2) and shall include the following:

(a) Task 102 of MIL-STD-470 and MIL-STD-785, monitor/control of subcontractors and vendors.
(b) Task 103 of MIL-STD-470 and MIL-STD-785 program reviews.
(c) Task 104 of MIL-STD-470 and MIL-STD-785, failure reporting, analysis, and corrective action system for reliability and a data collection, analysis and corrective action system for maintainability.
(d) Task 203 of MIL-STD-470 and MIL-STD-785, R and M predictions. The reliability prediction shall use the naval submarine environmental factors specified in MIL-HDBK-217. The prediction shall be performed using the part stress analysis method and the failure rates specified in MIL-HDBK-217. The maintainability prediction shall be performed in accordance with MIL-HDBK-472, and predictions for both primary and secondary maintenance as defined herein (see 3.11.3.1 and 3.11.3.2) shall be prepared in accordance with the data ordering documents included in the contract or order (see 6.2.2).
(e) Task 206 of MIL-STD-470, maintainability design criteria.
(f) Task 207 of MIL-STD-785, parts program in accordance with procedure 1 of MIL-STD-965.
(g) Task 301, of MIL-STD-785, environmental stress screening, shall be conducted when specified (see 6.2.1).

Power supplies contained within the equipment (plus 24 ± 15 Vdc) shall be subjected to workmanship screening as specified (see 6.2.1).

3.10.3 Design review. A formal reliability and maintainability design review program shall be established and conducted in accordance with the data ordering documents included in the contract or order (see 6.2.2). Participants shall include contractor personnel versed in design, reliability and maintainability. Naval Sea Systems Command (NAVSEA) representatives will have the option to attend all formal design reviews. NAVSEA shall be notified at least 15 days in advance of each review date. Notification shall include an agenda of the review including a check list of the design aspects to be covered. Reviews shall include:

(a) Reliability and maintainability (task 103 of MIL-STD-785 and MIL-STD-470).
(b) Design.
(c) Value engineering.

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(d) Safety engineering.
(e) Standardization of parts.
(f) Means of reducing maintenance.
(g) Improvement in methods and materials.
(h) Inspection procedures.

3.10.4 System safety program plan.

3.10.4.1 A system safety program plan shall be prepared for conducting a safety program in accordance with the data ordering documents included in the contract or order (see 6.2.2), and as specified herein.

3.10.4.2 System safety program plan shall delineate the equipment design techniques by which the contractor will comply with the safety requirements of MIL-E-917 and MIL-STD-454. MIL-STD-882 shall be used as guidance in the preparation of the system safety program plan.

3.11 Reliability and maintainability.

3.11.1 Definition of failure. For purposes of determining reliability, a failure shall be defined as any unscheduled shutdown of the power supply or performance which deviates from the specified operating limits stated in this specification.

3.11.2 Reliability requirements. Mean-time-between-failure of the power supply shall be not less than 10,000 hours. Unless otherwise specified herein, definition of terms for reliability shall be in accordance with MIL-STD-721.

3.11.3 Maintainability requirements. Maintainability requirements shall be equipment repair time (ERT) as defined in MIL-STD-471, method 3. ERT of the equipment for primary corrective maintenance shall be 0.4 hour. Maximum corrective maintenance (MCTMAX) for primary corrective maintenance shall be 0.75 hour. Definition of terms for maintainability shall be in accordance with MIL-STD-721.

3.11.3.1 Primary corrective maintenance. Equipment shall be designed so that primary corrective maintenance can be performed by replacement of subassemblies. In the event special tools are required for removal or insertion of subassemblies, such tools shall be supplied with the power supply. Special tools are defined as those tools not listed in the Federal Supply Catalog. Copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA). Equipment design shall conform to the design-for-maintainability requirements of MIL-STD-470 and NAVSEA 0967-LP-312-8010.

3.11.3.2 Secondary corrective maintenance. Subassemblies intended as replaceable items for primary corrective maintenance actions in case of equipment failure shall be designed to facilitate shipboard repair. Subassembly repairs requiring higher echelon corrective maintenance (as defined in NAVSEA 0967-LP-312-8010) shall be so identified in the applicable technical manual (see 3.10). Except for irreplaceable parts, replaceable subassemblies shall be capable of repair by ship's force through replacement of individual parts. Where special tools are required for subassembly repair, such tools shall be furnished with the power supply.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Unless otherwise specified herein, the examination and test listed in table IV shall be performed in accordance with MIL-P-15736. The examination and tests of table IV shall demonstrate that the power supply satisfies the requirements of this specification.

TABLE IV. First article and quality conformance inspection.

Examination and tests	First article inspection		Quality conformance inspection	Requirement reference	Test reference
General examination	X		X	3.2	4.5
Insulation resistance	X		X	MIL-P-15736	MIL-P-15736
Dielectric	X		X	MIL-P-15736	MIL-P-15736
	Oxygen generator load	Simulated oxygen generator load	Simulated oxygen generator load		
Current regulation	X	X	-	3.4	4.6.1
Phase balance	-	X	X	3.4	4.6.2
Harmonic distortion	X	X	X	3.5	4.6.3
Output ripple	X	X	X	3.6	4.6.3.1

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TABLE IV. First article and quality conformance inspection. - Continued

Examination and tests	First article inspection		Quality conformance inspection	Requirement reference	Test reference
Cooling	X	X	X[1/]	3.3.2	4.6.4
Protective shutdown	-	X	X	3.3.2	4.6.5
Airborne noise	X	X	X	3.7	4.6.6
Structure-borne noise	X	X	X	3.8	4.6.6
Shock	-	X	-	3.9	4.6.7
Vibration	-	X	-	3.9	4.6.7
Maintain-ability	-	X	-	3.11.3	4.6.8
Safety	-	X	X	3.3 and 3.10.4	4.6.9
Burn-in	-	X	-	3.2.7	4.6.10
Endurance	X	-	-	3.2.8	4.6.11

1/ Cooling test shall be performed on each tenth production unit of the power supply.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in table IV. Power supply shall be tested under oxygen generator load or simulated oxygen generator load as specified in table IV for each first article test. Testing with generator load is limited to 1050 amperes. Simulated oxygen generator load tests will be performed up to 1250 amperes. Unless otherwise specified (6.2.1), the oxygen generator described by NAVSEA S9515-AA-MMO-010/6L16, NAVSEA S9515-AA-MMO-020/6L16, and NAVSEA S9515-AA-MMO-030/6L16 shall be made available at a Government laboratory designated by the contracting activity (see 6.2.1). Oxygen generator will be made available and operated by Government personnel to support first article testing requiring an oxygen generator load. First article inspection and testing may be waived at the discretion of the contracting activity.

4.3.1 Government testing. The following first article tests will be performed by the Government at a Government laboratory:

(a) Current regulation.
(b) Harmonic distortion.
(c) Output ripple.
(d) Cooling.
(e) Protective shutdown.
(f) Airborne noise.
(g) Structureborne noise.
(h) Endurance.

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The Government shall require advance written notification to schedule the first article tests at the designated facility.

4.4 Quality conformance inspection. Each power supply offered for delivery shall be subjected to the examination and tests specified in table IV. Power supply shall be tested under resistance load during quality conformance inspection as specified in table IV.

4.5 General examination. Power supply shall be visually examined to determine that quality of workmanship conforms to MIL-E-917 and that equipment construction, as described by applicable engineering drawings, meets the requirements on interchangeability (see 3.2) and design (see 3.3), specified herein.

4.6 Tests.

4.6.1 Current regulation. Power supply shall be tested under oxygen generator load at each condition of input voltage, frequency, and ambient temperature specified in table V. Inlet cooling water temperature shall be 35°C. For each condition of input line voltage, frequency, and ambient temperature listed in table V, dc output current shall be adjusted as follows:

(a) Dc output current shall be adjusted at a setting of zero Adc in order to limit any surge of current when electrolysis is started.

(b) Output current adjustment setting shall be advanced from zero to 350 Adc. After start of electrolysis at 350 Adc output current setting shall be advanced in 100 ampere steps at 20-minute intervals until a setting of 1250 amperes is obtained.

Readings of input voltage, frequency, ambient temperature, dc output voltage and dc output current shall be recorded at each specified setting of dc output current. Curves shall be plotted of output current versus output voltage for each of the specified current regulation tests. Current regulation test shall demonstrate compliance with 3.4.

TABLE V. Conditions of line voltage, frequency, and ambient temperature.

Input voltage (volts)	Input frequency (Hz)	Ambient temperature (°C)
Rated	Rated	50
Rated + 5 percent	Rated	50
Rated + 5 percent	Rated	50
Rated - 5 percent	Rated	50
Rated - 5 percent	Rated	50
Rated + 5 percent	Rated	23
Rated + 5 percent	Rated	23
Rated - 5 percent	Rated	23
Rated - 5 percent	Rated	23
Rated	Rated	23

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4.6.2 Phase balance. Power supply shall be tested at 350 Adc and at 100 amperes steps up to 1250 Adc for current balance in the ac lines. Measurement of the rms value of ac line current of each phase shall be performed in order to demonstrate compliance with 3.4.

4.6.3 Harmonic distortion. Power supply equipment shall be tested to determine compliance with the limitations on allowable magnitude of harmonic line current specified in 3.5. The ac input line which feeds the power supply shall be monitored for distortion. Under the conditions of test, the power source shall not have a harmonic voltage content at any frequency in percent rms which is greater than 25 percent of the allowable percentage harmonic current at that frequency as specified in 3.5. Output load current shall be varied from 350 to 1250 Adc in steps of 100 amperes. At each load step commencing with 350 amperes, fundamental frequency component and harmonic components of input line current in lines A, B, and C of the 440-volt line shall be measured. Each harmonic line current of order 2 through 46 shall be recorded as rms amperes for each of lines A, B, and C. Harmonic line current shall be plotted against frequency. Either of the following methods of measurement of harmonic line current may be used:

(a) A noninductive line resistor shall be inserted in each input line. Fundamental frequency and harmonic line currents shall be measured by determining the voltage developed across the line resistor. The resistance value of the line resistor shall be such that the potential drop at fundamental frequency will be 1 percent of line voltage.
(b) Fundamental frequency and harmonic input line currents shall be measured by means of current probe.

Accuracy of measurement of harmonic currents shall be 5 percent. A wave analyzer shall be used for measurement of voltages resulting from sensing of line current.

4.6.3.1 Output ripple. Power supply shall be operated from the same power source as specified in 4.6.3. Input ac line voltages shall be balanced. Output load shall be varied from 350 to 1250 Adc in steps of 100 Adc. Output voltage and dc output current shall be recorded. Total rms current ripple shall be recorded at each load step. Individual frequency components of current ripple from the first through 333 harmonic of power line frequency shall also be recorded at each load step. Measurement of current ripple shall demonstrate conformance to 3.6.

4.6.4 Cooling.

4.6.4.1 Power supply shall be operated at 1250 Adc output current, under simulated oxygen generator load at 55 Vdc, with rated line input voltage applied. Inlet cooling water temperature shall be maintained at 35°C at ambient temperature of 50°C. Cooling water shall be supplied at 5 gal/min flow rate, at a maximum gauge pressure of 100 lb/in^2, with 20 lb/in^2 maximum pressure drop across the power supply. Purity of the cooling water shall be as specified in 3.3.2.2. Flow rate, pressure, temperature, and resistivity of inlet cooling water shall be recorded for test. Temperatures shall be monitored at selected points within the power supply equipment, particularly at transformer and reactor cores, winding

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and insulation surfaces, at power semiconductor studs and heat sinks, and at filter capacitor cases. For air cooled coils of transformers and reactors, the copper temperature of the coil shall be measured by the resistance method in accordance with MIL-E-917, and the temperatures recorded. The cooling test shall continue until component temperatures have stabilized. Temperature shall be considered stable when the rate of temperature change does not exceed 1°C per hour, as established by three successive readings of temperature recorded at 1-hour intervals. Heat run test data shall demonstrate that temperatures of wound magnetic devices conform to MIL-E-917, with respect to the assigned thermal classification of insulation systems and materials. Heating data shall also verify that all power semiconductors and filter capacitors are operated within temperature ratings defined by specification or manufacturer's product data. Cooling water pressure drop from inlet to outlet power supply, and outlet temperature shall also be recorded in accordance with 3.3.2. Measurement of pressure drop across the power supply shall demonstrate conformance to 3.3.2.

4.6.4.2 Power supply shall be installed in the oxygen generator described by NAVSEA S9515-AA-MMO-010, NAVSEA S9515-AA-MMO-020 and NAVSEA S9515-AA-MMO-030, and shall be operated at 1250 Adc output current under oxygen generator load with rated line input voltage applied. Cooling test shall be as specified in 4.6.4.1, with respect to cooling water, ambient temperature, and measurements of power supply temperature and pressure drop.

4.6.4.3 Quality conformance inspection. Power supply shall be operated at 1250 Adc under simulated oxygen generator load with rated line input voltage applied. Heat run shall be as specified in 4.6.4.1.

4.6.5 Protective shutdown. Cooling water supply shall be shut off with the power supply operating at 1250 Adc. Temperature of power semiconductors shall be recorded in accordance with the procedure of 4.6.4.1. This test shall demonstrate that the power supply is shut down by operation of internal protective circuitry (see 3.3.2) before maximum permissible temperature ratings of semiconductors are exceeded.

4.6.6 Airborne and structureborne noise.

4.6.6.1 First article noise tests. Two separate tests shall be performed, one with the power supply operated with simulated oxygen generator load to a maximum of 1250 amperes, as described herein. One with the power supply operated with an oxygen generator load to a maximum of 1250 amperes. The simulated oxygen generator load test shall be performed first and the power supply shall be mounted on a test fixture which is separate from the oxygen generator. The first article power supply shall then be operated with the oxygen generator as specified in NAVSEA S9515-AA-MMO-010, NAVSEA S9515-AA-MMO-020, NAVSEA S9515-AA-MMO-030, and NAVSEA S9515-AA-MMO-040. The power supply under test and the oxygen generator shall be separated a sufficient distance to ensure that ambient airborne noise at the power supply test fixture is at least 10 dB below the sound pressure acceptance levels specified in 3.7. Power supply shall be operated with output current initially set at zero. Output current adjustment setting shall be advanced from zero to 350 Adc. Then the output current setting shall be advanced to 100 ampere steps every 20 minutes (minimum) until a setting of 1250 amperes is obtained. The procedure for airborne noise test shall be in accordance with MIL-STD-740 for sound pressure level measurements. Structureborne noise test

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shall be performed and reported in accordance with MIL-STD-740. Sound pressure and vibrational acceleration shall be measured at each specified output current adjustment setting between 350 and 1250 amperes. Sound pressure level of power supply shall demonstrate conformance to 3.7. Structureborne noise test shall demonstrate conformance to 3.8.

4.6.6.2 For quality conformance testing of airborne and structureborne noise, the power supply shall be operated at 1250 Adc under simulated oxygen generator load. Procedure for airborne noise test shall be in accordance with MIL-STD-740 for sound pressure level measurements. Structureborne noise test shall be performed and reported in accordance with MIL-STD-740. Sound pressure level of the power supply shall demonstrate conformance to 3.7. Structureborne noise test shall demonstrate conformance to 3.8.

4.6.7 Shock and vibration. Power supply shall be subjected to shock and vibration tests as specified in MIL-P-15736 to determine conformance to 3.9. The power supply shall be energized at 350 Adc output current and the cooling water system pressurized to 100 lb/in^2.

4.6.8 Maintainability. Equipment malfunction shall be simulated in accordance with MIL-STD-471. Troubleshooting and repair procedures shall be performed in accordance with the applicable technical manual (see 3.10). Equipment repair shall be demonstrated in a manner consistent with MIL-STD-471. Examination of equipment maintainability shall verify compliance with the requirements of ERT and MCTMAX specified in 3.11.3.

4.6.9 Safety. Equipment shall be visually examined in order to determine that safety devices have been incorporated to provide for protection to personnel in accordance with safety requirements specified herein. Power supply shall also be operated under simulated oxygen generator load and the actuation of safety devices shall be demonstrated in such a manner as to verify conformance to this specification.

4.6.10 Burn-in. Power supply shall be operated at 1250 Adc under simulated oxygen generator load for a period of 504 hours. The following measurements shall be recorded at the beginning of, and at 50-hour intervals during the burn-in test:

(a) Input voltage (3 phases).
(b) Input current (3 phases).
(c) Input power (kW).
(d) Input line current harmonics (3 lines).
(e) Output current ripple.
(f) Output current.
(g) Output voltage.
(h) Inlet and outlet water pressure.
(i) Inlet and outlet water temperature.
(j) Cooling water flow rate.

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Air ambient temperature shall be maintained at 50°C. Cooling water shall be supplied at temperature of 35°C, with flow rate of 5 gal/min. In the event of failures during test, the failed part shall be replaced and the test shall be continued. Cause of component failure and corrective action shall be reported to the contracting activity (see 4.7). Burn-in test shall be performed prior to the endurance test.

4.6.11 Endurance. Power supply shall be installed in the oxygen generator described by NASVEA S9515-AA-MMO-010, NAVSEA S9515-AA-MMO-020, and NAVSEA S9515-AA-MMO-030, and operated at rated full-load to furnish power to the electrolytic cells of the oxygen generator for a period of 336 hours (14 days). The following performance parameters of the power supply shall be monitored at the beginning of the test and at each interval of 112 hours thereafter until completion of test:

(a) Temperature rise of cooling water, to demonstrate compliance with 3.3.2.
(b) Input rms line current in each phase, to demonstrate compliance with 3.4.
(c) Dc output current.
(d) Fifth and seventh order input harmonic line currents, to demonstrate compliance with 3.5.
(e) Fundamental frequency component of output ripple current, to demonstrate compliance with 3.6.

Cooling water resistivity shall be monitored at beginning and end of test. Oxygen gas production and cell temperature of the electrolytic cells of the oxygen generator shall also be monitored during test. In the event of power supply failure or any shutdown of the oxygen generator prior to completion of the 336-hour operating period, the endurance test shall be discontinued. Cause of power supply failure or shutdown of the oxygen generator shall be reported to the contracting activity, with recommendations on corrective action or equipment design changes necessary to restore the oxygen generator to operation. The endurance test shall not be resumed until the contracting activity has furnished instructions on continuation of test.

4.6.12 Post installation test. Power supply should be installed on board within oxygen generator and post installation testing performed in accordance with NAVSEA S9515-AA-MMO-040.

4.7 Test procedures and test reports. First article test procedures, test reports and quality conformance test reports shall be prepared in accordance with the data ordering documents included in the contract or order (see 6.2.2).

4.8 Parts control program. The contractor shall maintain a parts control program in accordance with MIL-STD-965. Requirements for parts selection shall be as specified in MIL-STD-965. NAVSEA 0967-LP-597-1011 shall be used for parts derating.

4.9 Inspection of packaging. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation-packaging, packing, and marking.

5.1.1 Equipment. Each equipment with accessories shall be preserved-packaged level A or C, packed level A, B, or C as specified (see 6.2.1), and marked in accordance with MIL-E-15298.

5.1.2 Semiconductors or solid state devices. Semiconductors such as diodes, transistors, integrated circuits as well as circuit boards or chassis in which they are incorporated, shall be individually packaged in a barrier bag conforming to class E, style 1, type I or II, or class F, style 1, type I of MIL-B-117. MIL-B-117 bag material shall employ aluminum foil as a laminate of the bag barrier material. Bag closure shall be affected by heat sealing. Leads and terminals shall be protected from damage by means of the container (carrier) design, die cut inserts, or by the use of noncorrosive cushioning material. Leads and other projecting parts may be used for positioning, but shall not be subjected to loads or other stresses such as bending or twisting that can damage the entry seals. For level C preservation-packaging, semiconductors or solid state devices subject to electromagnetic degradation shall be protected with a wrap of aluminum foil or a barrier bag employing aluminum foil as a laminate of the bag material. Leads and terminals shall be protected as specified herein.

5.1.3 Cushioning, dunnage, and wrapping materials.

5.1.3.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill material for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for materials destined for shipboard installation/stowage.

5.1.3.2 Level C preservation-packaging and packing. When loose-fill type materials are used for packaging and packing applications such as cushioning, filler, and dunnage, all containers (unit, intermediate, and shipping) shall be marked or labeled with the following information:

"CAUTION

Contents cushioned with loose-fill material shall not be taken on board ship. Remove and discard loose-fill material before shipboard stowage. If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material."

5.2 Special requirements. Equipment shipping containers (see 5.1.1) shall be stenciled, and furnished with unpacking instructions, as specified in MIL-STD-740.

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5.3 Technical manuals. Technical manuals, which accompany shipments that are packed level A or B, shall be packaged in a transparent waterproof plastic bag, minimum 4 mil thick. Closure shall be by heat sealing. Technical manuals shall not be placed within any flexible sealed barrier enclosing components. The copies of the manual shall be placed in the shipping container housing the main unit. Packing lists shall indicate which container contains the technical manual(s) and shall also state the approximate location therein. For ease of removability the location of the manual(s) shall be such that it is readily accessible when the container is opened. Technical manuals, when shipped in bulk quantities, shall not be individully wrapped, but shall be packed in accordance with the requirements of the applicable technical manual specification or packed in containers conforming to the requirements for level A, B, or C.

5.4 Data (see 3.10). Data shall be prepared for delivery in such a manner as to ensure the required information is protected against deterioration, physical damage, or loss during shipment from the contractor to the receiving activity. Packages or shipping containers shall, as a minimum, conform to the level C requirements of 5.1.1.

6. NOTES

6.1 Intended use. This specification covers power supplies intended for use in oxygen generators on board submarines.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article inspection, if required (see 3.1).
(c) If environmental stress screening is to be conducted and what workmanship screening the power supplies are to be subjected to (see 3.10.2).
(d) Government laboratory designated to perform first article tests which require the power supply to operate with oxygen generator load (see 4.3).
(e) Levels of preservation-packaging and packing required (see 5.1.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

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Paragraph no.	Data requirement title	Applicable DID no.	Option
3.10	Drawings, engineering and associated lists	DI-E-7031	Level 3 Design activity designation - contractor Drawing no. - contractor
3.10.2	Reliability program plan	DI-R-7079	---
3.10.2	Maintainability program plan	DI-R-7103	---
3.10.2	Reliability prediction report	DI-R-7082	---
3.10.2	Maintainability predictions report	DI-R-7108	---
3.10.3	Conference agenda	DI-A-7088	---
3.10.4.1	System safety program plan	DI-H-7047	---
4.7	Test procedures (first article)	UDI-T-23732	---
4.7	Report, operational/ performance test (first article)	UDI-T-23740	---
4.7	Report, operational/ performance test (quality conformance)	UDI-T-23740	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article. When a first article inspection is required, the item should be a first article sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.3.1 Retest of first article sample. If the first article sample is disapproved by the Government, the contractor may elect, with the concurrence of the Government, to repeat any or all of the first article tests. After each notification by the Government of the requirement for additional tests, the contractor will, at no additional cost to the Government, make any necessary changes, modifications, or repairs to the first article sample or select another first article sample for testing. Thereafter, the contractor will perform the required additional tests and deliver another report to the Government under the terms and conditions and within the time specified by the Government.

6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6130-N244)

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